# AutoResearch: Preserving Source Status in Manuscript Claims under Incomplete Experimental Evidence

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**Abstract**

Agent-assisted drafting can detach manuscript claims from experiment records whose status remains unresolved. AutoResearch addresses that failure mode by binding each reportable result to an evidence unit: protocol, artifact, source status, allowed use, and closure condition. The evaluation uses PHMGA, a fault-diagnosis case with uneven evidence across a synthetic/offline proxy, an eight-row Stage B backend comparison, and an RM101 mixed-domain simulated-planner analysis. The synthetic/offline proxy reaches 1.0000 macro-F1 but remains below benchmark evidence. Stage B contains two accepted Ottawa rows, three rejected rows, and three pending rows; the Ottawa Codex row reaches 0.8775 macro-F1, matching the accepted OpenRouter/Nemotron Ottawa row, but remains ineligible for selection because planning and reflection used a fallback provider path. The RM101 simulated planner raises macro-F1 from a 0.8229 real baseline to 0.8865, yet remains mechanism-level evidence because the planner is simulated and the split is mixed-domain. These records support one methodological claim: source status remains attached to claims where proxy promotion, authority inversion, and mechanism inflation can arise. They do not support selected-backend readiness, cross-dataset performance, or a final diagnostic performance claim.

## 1 Introduction

Agent-assisted research workflows create a distinct reporting problem at the point where result records become manuscript claims. An experiment can remain unresolved in its source artifacts while its numerical output is written as if the claim were settled. Reporting standards already motivate transparent accounts of data, code, materials, and protocols, and publication policies emphasize originality, disclosure, and peer-review accountability [1, 2]. The narrower risk addressed here is loss of source status during drafting. AutoResearch attaches each substantive statement to an evidence unit, so the manuscript records whether a result is accepted, rejected, pending, proxy, or mechanism-level evidence before the claim is written.

Prior language-agent work addresses upstream capabilities. ReAct studies interleaved reasoning and action [3]; Reflexion studies verbal feedback across trials [4]; and Toolformer studies when a model should invoke external tools [5]. AutoResearch starts after candidate artifacts exist. Its object of control is not action selection, feedback use, or tool invocation, but the admissibility of manuscript claims made from unfinished result records. That placement keeps negative results, pending rows, and closure conditions attached to the statements they constrain.

AutoResearch is evaluated as a reporting method, not as another reasoning agent or a completed diagnostic model. Its contribution is to keep source status with the claim it licenses: accepted, rejected, pending, proxy, and mechanism-level results each restrict what can be said from them. The evaluation uses three claim-promotion errors. Proxy promotion writes a synthetic or offline signal as benchmark evidence; authority inversion lets metric rank override source status; and mechanism inflation writes a simulated or mixed-domain signal as online or out-of-domain performance. The synthetic/offline material, the PHMGA Stage B comparison, and the RM101 simulated-planner analysis test those errors.

| Source status | Allowed manuscript use | Claim still withheld |
|---|---|---|
| Accepted | Report the result within the documented protocol, dataset, and review scope. | Generalization beyond the accepted protocol. |
| Rejected | Report the row as negative evidence and keep the failure reason visible. | Success, backend selection, or repaired performance. |
| Pending | Report a metric only with the unresolved condition attached. | Ranking, selection, or closure based on that metric. |
| Proxy | Use the result as a sanity check or motivation for formal testing. | Benchmark-level or real-data performance. |
| Mechanism-level | Use simulated or bounded evidence as a hypothesis about model structure. | Online performance, out-of-domain generalization, or statistical reliability. |

Table 1: Evidence-status policy. The table defines the strongest manuscript use allowed by each source status before numerical results are interpreted.

PHMGA is the evaluation case because its record is uneven in ways that create those errors. It contains traceable synthetic/offline material, two accepted Ottawa Stage B rows, rejected RM101 rows, an Ottawa Codex v3 row that reaches 0.8775 macro-F1 but remains pending after provider fallback, and an RM101 mixed-domain DAG signal whose strongest simulated row reaches 0.8865 macro-F1 against a 0.8229 real baseline. The contrast places high metrics next to unresolved source conditions. The test is whether those conditions remain visible in prose, tables, and captions; selected-backend readiness and cross-dataset diagnostic performance remain outside the claim.

## 2 Evidence Model

An evidence unit defines when an experimental result can be used in manuscript text. It pairs the measured result with the protocol that produced it, the artifact inspected, the source status assigned to it, the claim boundary, and the condition that would close that boundary. The unit is needed where a result becomes prose, a table entry, or a caption, because those are the places where accepted, rejected, pending, proxy, and mechanism-level evidence can be made to sound stronger than its source allows.

The status is read before the metric. Accepted units support claims only within their documented scope. Rejected rows remain negative evidence. Pending rows may report metrics, but they cannot decide a backend. Proxy or mechanism-level units can motivate further testing, but they cannot become performance evidence. The evidence model asks for the strongest statement source status permits, not the largest number in the row.

Table 1 states the policy used to read the PHMGA case.

PHMGA then supplies an uneven case. The synthetic/offline result has a perfect single-run score but remains proxy evidence. Stage B contains accepted Ottawa evidence, rejected RM101 evidence, and a high-scoring Ottawa Codex v3 row left pending by provider fallback. The RM101 all-domain mixed case contains a simulated-planner signal. These sources motivate the source-status test, but none establishes cross-dataset generalization, selected-backend readiness, Stage C/D completion, or a final diagnostic performance claim.

## 3 Methods

PHMGA is analyzed as an uneven evidence set rather than as a completed benchmark. The record contains accepted Ottawa rows, rejected RM101 rows, pending backend rows, and a

| Evidence row | Test accuracy | Test macro-F1 |
|---|---|---|
| Baseline pipeline | 0.8333 | 0.8286 |
| AutoResearch run | 1.0000 | 1.0000 |

Table 2: Preliminary synthetic/offline signal. The table reports one offline run only; it includes no variance, no real-data row, no RM101 resolution, and no selected-backend decision.

simulated RM101 mechanism signal. That mix creates the condition of interest: a result may carry a strong metric while the admissible claim remains narrow. The analysis starts from evidence units, each defined by the protocol, expected output, source artifact, status, and review or approval state available before interpretation.

Interpretation follows source status before score magnitude. Accepted, rejected, pending, proxy, and mechanism-level sources inherit the limits in Table 1; a prose claim, table entry, or caption can be no stronger than that status permits. When a stronger reading is possible but not yet licensed, it is recorded as a withheld claim together with the evidence that would be needed to make it.

Before a result is used, the manuscript asks four questions: what was measured, which source verdict is attached, what claim that verdict permits, and what evidence would be required to go further. A stream supports the method only when those answers remain visible in the sentence, table, or caption that reports the result. A high metric can then be reported without being used as backend ranking, selection, or performance evidence by itself.

The evaluation then checks three claim-promotion errors. Proxy promotion occurs when a synthetic or offline signal is written as benchmark or real-data evidence. Authority inversion occurs when metric rank overrides source status, for example when a pending or rejected row is used for backend selection. Mechanism inflation occurs when a simulated or mixed-domain signal is written as online or out-of-domain performance. In PHMGA, the synthetic/offline stream tests proxy promotion, the Stage B stream tests authority inversion, and the RM101 stream tests mechanism inflation.

Closure is assigned by the source verdict or approval record, not by the fluency of the paragraph that reports it. For reproducibility, each reported result keeps the identifiers needed to reconstruct provenance: protocol, source record, source artifact, evaluator, result schema, and model/provider metadata when agents are used.

# 4 Results

The three evidence streams are organized by the claim upgrade each could invite. The synthetic/offline stream is the proxy case. Under the same Ottawa-style offline condition, the AutoResearch run records higher accuracy and macro-F1 than the baseline pipeline (Table 2). The status assigned to the row stays with the metric: the 1.0000 macro-F1 remains a proxy, not benchmark evidence.

The 1.0000 synthetic score is reported as evidence for status retention, not performance. It supports the narrower observation that a strong offline signal can be carried forward without authorizing a stronger empirical claim. The missing evidence remains visible: no accepted Stage C/D formal row, no repeated-run variance, no RM101 resolution, and no selected backend.

The PHMGA Stage B comparison provides the real-data check for authority inversion. Table 3 reports the metric-bearing rows. The accepted Ottawa evidence is narrow: OpenRouter/Nemotron reaches 0.8775 macro-F1 and BigModel GLM-4.7 reaches 0.7227 under Ottawa only. The corresponding RM101 evidence is negative, with OpenRouter/Nemotron at 0.1834 macro-F1 and BigModel GLM-4.7 at 0.1893. A later Ottawa Codex v3 run emitted the same 0.8775 macro-F1 as the accepted OpenRouter/Nemotron row, but its planning and

| Dataset | Backend | Evidence status | Evidence note | Test macro-F1 |
|---|---|---|---|---|
| Ottawa | OpenRouter/Nemotron | accepted | checks passed | 0.8775 |
| Ottawa | BigModel GLM-4.7 | accepted | checks passed | 0.7227 |
| Ottawa | Codex v3 | pending | fallback path | 0.8775 |
| RM101 | OpenRouter/Nemotron | rejected | plan revision needed | 0.1834 |
| RM101 | BigModel GLM-4.7 | rejected | implementation revision needed | 0.1893 |

Table 3: PHMGA Stage B real-data rows. The Codex row is reported only as pending evidence because the provider path fell back. The table does not select a global backend because no backend has accepted evidence on both Ottawa and RM101, and Stage C/D rows remain unavailable.

| Row | Planner status | Test accuracy | Test macro-F1 | Interpretation |
|---|---|---|---|---|
| BigModel GLM-4.7-FlashX | real baseline | 0.8141 | 0.8229 | reference baseline only |
| Gemini 2.0 Flash | simulated variant | 0.1539 | 0.1305 | negative simulated variant |
| Gemini 2.5 Flash | simulated variant | 0.2585 | 0.2525 | negative simulated variant |
| Gemini 2.5 Pro | simulated variant | 0.8865 | 0.8865 | mechanism-level DAG signal |
| Cross-DAG late fusion | post hoc fusion | 0.8241 | 0.8299 | below best single DAG |

Table 4: RM101 all-domain mixed case study. The strongest row is a simulated planner variant; this table does not resolve the RM101 Stage B rejection, support online Gemini performance, or establish leave-domain-out generalization.

reflection used a fallback provider path. The highest visible score is not the strongest evidence.

Across the complete active Stage B set, eight rows resolve to two accepted Ottawa rows, three rejected rows, and three pending rows. Table 3 shows the subset with direct metric evidence. The interpretation follows status before score: accepted source status authorizes an Ottawa-only report, rejected RM101 rows remain negative evidence, and pending rows identify closure tasks rather than backend winners. This pattern is the real-data evidence for the authority-inversion part of source-status preservation.

The RM101 all-domain mixed experiment is separate from the Stage B acceptance decision. Its role is narrower: it tests whether richer, physics-aware planner DAGs produce a measurable signal under a fixed RM101 mixed-domain protocol. The protocol uses `Dataset_id=101`, all 12 domains, split-before-windowing stratification, window size 4096, stride 4096, 146/46/48 source records, and 27448/8648/9024 train/validation/test windows. Table 4 reports the result boundary.

The RM101 signal is not a general gain from simulated planning. Two simulated variants remain far below the real baseline, and cross-DAG late fusion stays near the baseline. The only stronger row is a specific operator path: short-time Fourier structure, RMS aggregation, and torque normalization on a gearbox vibration channel. That pattern gives a concrete branch for the next RM101 run, while the simulated planner status and mixed-domain split keep the claim below online performance or leave-domain-out generalization.

Across the three streams, the supported result is narrow: source status remains attached while claims are formed. The synthetic row keeps a perfect offline score below benchmark strength, blocking proxy promotion. Stage B keeps the high-scoring pending Codex row outside backend selection while rejected RM101 rows remain visible, blocking authority inversion. The RM101 case keeps the physics-aware DAG branch as a target for a formal RM101 run, blocking mechanism inflation. A diagnostic performance claim still requires one backend accepted on both datasets, accepted Stage C and Stage D rows, and repeated real-data runs with variance.

| Observed result | Source-status decision | Claim withheld | Evidence still needed |
|---|---|---|---|
| Synthetic/offline sanity check improves from 0.8286 to 1.0000 macro-F1 in one offline run. | The proxy result stays below benchmark strength; proxy promotion is blocked. | Benchmark-level or real-data improvement. | Repeated real-data runs with variance. |
| Stage B rows include accepted Ottawa scores of 0.8775 and 0.7227, rejected RM101 scores of 0.1834 and 0.1893, and a pending Ottawa Codex score of 0.8775. | Backend selection follows source status rather than metric rank; authority inversion is blocked. | Backend ranking or selected-backend claim from the high Codex number. | Backend accepted on both Ottawa and RM101. |
| RM101 mixed-domain case records a real baseline at 0.8229 macro-F1 and a best simulated planner row at 0.8865. | The physics-aware DAG signal remains a simulated, mixed-domain mechanism case; mechanism inflation is blocked. | Online RM101 performance, Stage B repair, or leave-domain-out generalization. | Accepted online backend and out-of-domain validation. |

Table 5: Claim-boundary synthesis. Each empirical stream states the source-status decision, the stronger claim still withheld, and the evidence needed before that claim could be made.

# 5 Discussion

Stage B does not identify a winning backend; it preserves the conflict between score and admissible status. Ottawa Codex v3 reports a 0.8775 macro-F1, the same value as the accepted OpenRouter/Nemotron Ottawa row. A score-only reading would make that row selection-relevant. Source status prevents that move: planning and reflection followed a fallback provider path, so the row remains pending and selection-ineligible while the accepted Ottawa rows and rejected RM101 rows remain visible. The high metric, failed provenance condition, positive Ottawa evidence, and negative RM101 evidence stay in one account without becoming backend selection evidence.

The RM101 simulated-planner case tests a different claim boundary. It does not repair the rejected RM101 Stage B rows; it identifies a physics-aware DAG branch for a formal RM101 run. The evidence is comparative within the bounded case: two simulated variants are poor, late fusion does not exceed the best single branch, and only the path that combines time-frequency structure with load normalization rises above the real baseline. That pattern supports a mechanism hypothesis rather than online diagnostic-performance evidence, because the planner is simulated and the split is mixed-domain.

Table 5 summarizes the result-to-claim test. Each row states the source-status decision, the stronger claim withheld, and the evidence needed before that stronger claim could be made.

The comparison matters because the source record, not the narrative summary, controls claim upgrades. A score-only summary could drop the pending Codex row, soften the rejected RM101 rows into a qualifier, or present the RM101 simulated result as a score alone. AutoResearch keeps the metric, source status, withheld claim, and closure condition in the same record. A clean provider rerun, accepted RM101 backend row, accepted Stage C/D rows, or repeated-run variance estimate can change the manuscript only after the corresponding source record changes.

The retained boundaries also preserve alternative explanations. The Ottawa rows may reflect dataset-specific separability rather than a portable backend advantage. The Codex row may reflect the fallback provider path rather than model quality. The RM101 simulated-planner signal may reflect mixed-domain structure rather than leave-domain-out generalization. Those explanations remain attached to withheld claims and closure conditions, rather than being folded into a performance narrative.

This leaves a defined manuscript boundary. Backend selection requires accepted Ottawa–RM101 evidence for the same backend, not an Ottawa-only score. A performance-centered claim requires accepted Stage C main-result rows, Stage D ablations, and repeated real-data estimates. Until those sources exist, the evidence supports source-status preservation and a bounded mechanism hypothesis; cross-dataset improvement, selected-backend readiness, statistical significance, and final diagnostic performance claims remain unclaimed.

## 6 Limitations

PHMGA's real metrics create the main scope risk. The accepted Ottawa rows, the high-scoring pending Codex row, and the stronger RM101 simulated-planner row can look like a partial diagnostic-model comparison. Their role here is different: they show whether source-status labels stay attached when metrics are reportable. The supported contribution is source-status preservation under incomplete evidence, not a diagnostic-model result.

The empirical base remains thin. The synthetic/offline result is one run, the Ottawa Codex row may describe a fallback provider path rather than the configured backend, and the RM101 planner rows are simulated. These facts explain why narrower claims are reported. They leave accepted online backend runs, repeated trials, confidence intervals, and variance estimates as open requirements.

Ottawa and RM101 also differ in status. Ottawa contains accepted Stage B rows, whereas the corresponding RM101 rows remain rejected. The RM101 mixed-domain split may reflect within-domain structure rather than leave-domain-out transfer. The simulated planner identifies a candidate branch for a formal RM101 run; it does not forecast that such a run will pass.

Evidence for a final diagnostic performance claim is still absent. No backend has accepted evidence on both datasets, Stage C/D rows are unavailable, and repeated real-data variance is absent. The present record supports evidence-unit contracts for source-status preservation and a physics-aware DAG candidate for RM101 testing. It does not support selected-backend readiness, cross-dataset improvement, statistical reliability, or a final diagnostic performance claim.

## 7 Conclusion

The PHMGA case leaves a narrow method result. AutoResearch keeps source status attached when metrics are carried into manuscript claims: the synthetic/offline 1.0000 macro-F1 score remains a proxy, the Ottawa Codex 0.8775 macro-F1 row remains pending after provider fallback, and the RM101 simulated-planner gain remains a mechanism hypothesis. These outcomes block proxy promotion, authority inversion, and mechanism inflation without erasing the metrics that motivate follow-up experiments. Claims about backend readiness or diagnostic performance require one backend accepted on both Ottawa and RM101, accepted Stage C/D rows, and repeated real-data runs with variance estimates.

## Data availability

The tables report derived metrics rather than redistributed raw data. Raw external or private datasets are not included. The reported values are traceable to the synthetic/offline sanity-check record, PHMGA Stage B backend-comparison records, and RM101 mixed-domain simulated-planner metadata. The accompanying metadata are limited to the fields needed to check source status, protocol scope, and the claim boundaries stated in the manuscript.

## Code availability

The accompanying materials include the manuscript source, validation scripts, table source files, and PHMGA evidence records used for the reported values. Claim evidence is limited to those table sources, validation scripts, and evidence records. Planning notes, display views, and repository-operation files are excluded from the manuscript evidence base.

## References

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